Q1) Find the Output

```
class A
{
   A()
   {
      System.out.println("A Constructor");
   }
}

class B extends A
{
   B()
   {
      System.out.println("B Constructor");
   }
}

public class Test
{
   public static void main(String[] args)
   {
      new B();
   }
}
```

Q2) Find the Output

```
class Alpha
{
   Beta b = new Beta();

   public Alpha()
   {
     System.out.println("Alpha class constructor");
   }


   {
     System.out.println("Non static block of Alpha class");
   }
}
class Beta
```

```
{
public Beta()
{
System.out.println("Beta class constructor");
}
}

public class Test
{
   public static void main(String[] args)
   {
      new Alpha();

   }
}
```

Q3) Find the Output

```
class ConstructorsTest
{
   public static void main(String[] args)
   {
      U u = new U();
      U u1 = new U(1);
      V v = new V();
      V v1 = new V(1);
      T t = new T();
      T t1 = new T(1);
   }
}
class U {
   V v = new V();

   U()
   {
    System.out.print("U");
   }

   U(int i)
   {
    System.out.print("u");
   }
}
```

```
class V
{
    V()
    {
     System.out.print("V");
    }

    V(int i)
    {
     System.out.print("v");
    }
}

class T extends V
{
    T()
    {
     System.out.print("T");
    }

    T(int i)
    {
     System.out.print("t");
    }
}
```

Q4) Find the Output

```
class OutPut
{
    public static void main(String[] args)
    {
        new Alpha();
        new Base();
    }
}
class Alpha extends Base
{
    Alpha()
    {
        System.out.print("Base");
    }
```

```
}
class Base
{
   Base()
   {
      System.out.println("Base");
   }
}
```

Q5) Find the Output

```
class Parent
{
   int x = 10;
}

class Child extends Parent
{
   int x = 20;
   void print()
   {
      System.out.println(x);
   }
}

public class Test
{
   public static void main(String[] args)
   {
      new Child().print();
   }
}
```

Q6) Find the Output

```
class A
{
   A()
   {
      System.out.println("A constructor");
   }
}
```

```
class B extends A
{
    B()
    {
        this(5);
        System.out.println("B no-arg constructor");
    }

    B(int x)
    {
        super();
        System.out.println("B int constructor: " + x);
    }
}

public class Test
{
    public static void main(String[] args)
    {
        new B();
    }
}
```

Q7) Find the Output

```
class Parent
{
    Parent(String msg)
    {
        System.out.println("Parent: " + msg);
    }
}

class Child extends Parent
{
    Child()
    {
        this("Hello");
        System.out.println("Child default");
    }

    Child(String s)
    {
```

```
        super(s);
        System.out.println("Child: " + s);
    }
}

public class Test
{
    public static void main(String[] args)
    {
        new Child();
    }
}
```

Q8) Find the Output

```
class A
{
    A()
    {
        this(10);
        System.out.println("A no-arg");
    }

    A(int x)
    {
        System.out.println("A int: " + x);
    }
}

public class Test
{
    public static void main(String[] args)
    {
        new A();
    }
}
```

Q9) What will be the output of the following program?

```
class Base
{
    Base()
    {
```

```
        System.out.println("Base constructor");
    }
}

class Derived extends Base
{
    Derived()
    {
        super();
        this(5);
        System.out.println("Derived no-arg");
    }

    Derived(int x)
    {
        System.out.println("Derived int: " + x);
    }
}

public class Test
{
    public static void main(String[] args)
    {
        new Derived();
    }
}
```

Q10) What will be the output of the following program?

```
class OutPut
{
    public static void main(String s[])
    {
        int i = 2;
        B b = new B();
        b.execute();
    }
}

class A
{
    A()
    {
```

```
        int i = 1;
    }
}

class B extends A
{
    void execute()
    {
        System.out.println("i = " + i);
    }
}
```

Q11) What will be the output of the following program?

```
class FourWheeler
{
    public static void main(String[] args)
    {
        new Car(0);
    }
}

class Vehicle
{
    int i;
    int j;

    Vehicle(int i)
    {
        this.i = i;
        j = i;
    }

    Vehicle(int i, int j)
    {
        this.i = i;
        this.j = j;
    }
}

class Bus extends Vehicle
{
    double j;
```

```
    Bus(double j)
    {
        this.j = j;
    }
}

class Car extends Vehicle
{
    Car(int i)
    {
        super(i);
        System.out.print("Executed Successfully");
    }
}
```

Q12) What will be the output of the following program?

```
class BaseClass
{
    BaseClass()
    {
        System.out.print("*");
    }
    BaseClass(String str)
    {
        this();
        System.out.print("@" + str);
    }
}
public class ChildClass extends BaseClass
{
    ChildClass()
    {
        System.out.print("#");
    }
    ChildClass(String name)
    {
        this();
        System.out.print("$" + name);
    }
    public static void main(String args[])
    {
```

```
        new ChildClass("hello");
    }
}
```

Q13) What will be the output of the following program?

```
class Base
{
    Base()
    {
        System.out.println("Base()");
    }
    Base(int x)
    {
        System.out.println("Base(int): " + x);
    }
}

class Derived extends Base
{
    Derived()
    {
        this(20);
        System.out.println("Derived()");
    }
    Derived(int y)
    {
        super(y);
        System.out.println("Derived(int): " + y);
    }
}

public class Test
{
    public static void main(String[] args)
    {
        new Derived();
    }
}
```

Q14) What will be the output of the following program?

```
class X
```

```
{
   X()
   {
      this(10);
   }
   X(int a)
   {
      this();
   }
}
public class Test
{
   public static void main(String[] args)
   {
      new X();
   }
}
```

Q15) What will be the output of the following program?

```
class Alpha
{
        public Alpha(String msg)
        {
                System.out.println("Alpha "+msg);
        }
}
class Beta extends Alpha
{
        public Beta()
        {
                super(getMessage());
                System.out.println("Beta class constructor!!");
        }

        public String getMessage()
        {
                return "class constructor";
        }
}

public class Test
{
```

```
  public static void main(String[] argv)
  {
        new Beta();
  }
}
```

Q16) What will be the output of the following program?

```
class Alpha
{
        public Alpha(String msg)
        {
                System.out.println("Alpha "+msg);
        }
}
class Beta extends Alpha
{
        public Beta()
        {
                super(getMessage());
                System.out.println("Beta class constructor!!");
        }

        public static String getMessage()
        {
                return "class constructor";
        }
}

public class Test
{
  public static void main(String[] argv)
  {
        new Beta();
  }
}
```

Q17) What will be the output of the following program?

```
class Alpha
```

```
{
        private Alpha(String msg)
        {
                System.out.println("Alpha "+msg);
        }
}
class Beta extends Alpha
{
        public Beta()
        {
                super("class constructor");
                System.out.println("Beta class constructor!!");
        }
}
public class Test
{
  public static void main(String[] argv)
  {
          new Beta();
  }
}
```

Q18) What will be the output of the following program?

```
package com.ravi.oop;

class Base
{
      public Base(String s)
      {
        System.out.println("Base class parameterized String");
      }
      public Base(int i)
      {
            System.out.println("Base class parameterized int");
      }
}
class Derived extends Base
{
      public Derived(String s)
      {
        super(s);
        System.out.println("Derived class parameterized String");
```

```
    }
    public Derived(int i)
    {
        super(i);
       System.out.println("Derived class parameterized int");
    }
}
public class Test
{
  public static void main(String[] argv)
  {
          new Derived("Java");
          //new Derived(10);
  }
}
```

Q19) What will be the output of the following program?

```
class A
{
int i;
void display()
{
System.out.println(i);
}
}
class B extends A
{
int j;
void display()
{
System.out.println(j);
}
}
class Inheritance
{
public static void main(String args[])
{
B obj = new B();
obj.i=1;
obj.j=2;
obj.display();
}
```

```
}
```

Q20) What will be the output of the following program?

```
class Constructors
{
   public static void main(String[] args)
   {
      new A(3);
      new B(5, 6);
      new B(6, 7).C(1, 2, 3);
      new C(3, 3, 4).B(4, 5);
      new C(1, 1, 1);
   }
}
class A
{
   A(int i)
   {
     print(i * 2);
   }
   void print(int i)
   {
   System.out.print(i + " ");
   }
}
class B extends A
{
   B(int j, int k)
   {
      super(j);
      print(k * 4);
   }
   void C(int j, int k, int l)
   {
     print(j + k + l - 3);
   }
}
class C extends B
{
   C(int j, int k, int l)
   {
      super(j, k);
```

```
        print(I - 3);
    }
    void B(int i, int k)
    {
      print(i * 6);
    }
}
```